



Disturbance observer based hierarchical control of coaxial-rotor UAV

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ABSTRACT

This paper propose an hierarchical controller based on a new disturbance observer with finite time convergence (FTDO) to solve the path tracking of a small coaxial-rotor-typs Unmanned Aerial Vehicles (UAVs) despite of unknown aerodynamic efforts. The hierarchical control technique is used to separate the flight control problem into an inner loop that controls attitude and an outer loop that controls the thrust force acting on the vehicle. The new disturbance observer with finite time convergence is intergated to online estimate the unknown uncertainties and disturbances and to actively compensate them in finite time. The analysis further extends to the design of a control law that takes the disturbance estimation procedure into account. Numerical simulations are carried out to demonstrate the efficiency of the proposed control strategy.

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1. Introduction

Unmanned aerial vehicles (UAVs) are promising to achieve many useful applications in both civil and military scenarios. The development of such systems pose a number of problems in sensing and control. We are particularly interested in controlling the coaxial rotor, a helicopter having two main rotors driven by two brushless DC motors rotating in opposite direction in order to overcome gyroscopic torque and cancel the drag torque produced by each rotor. The altitude is regulated by increasing or decreasing the thrust of both rotors. This helicopter uses a mechanical device known as a swashplate (collective pitch) driven by two servo motors placed on the lower rotor to change the pitch and roll angle incidence to obtain the pitch and roll control torques of the vehicle [24]. The yaw torque is ensured by a differential speed variation between the two rotors. A coaxial rotor is controlled by varying the angular speed of each rotor and the swashplate angle incidence of the lower rotor [2,3]. The force produced by each motor is proportional to the square of the angular speed. The coaxial system is under-actuated with only the 1-degree-of-freedom (DOF) thrust force input for the 3- dimensional Cartesian dynamics, although the rotational dynamics in SO(3) is fully-actuated.

Numerous strong control techniques have been proposed for the

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control of coaxial-rotor or similar systems. Many works focus on linear controllers have been developed to achieve flight performances of rotorcraft configurations like PD controller [5] and PID controller [6]. More recently, interest has been focused on nonlinear control laws like adaptive backstepping control in [7,8], nonlinear PID controller in [9], an integral predictive controller H_∞ [10], the control via singular perturbations [11]. In [12] an experimental vision regulation was applied to the quadrotor systems. A sliding mode controllers was presented in [2,13], a disturbance observer based control was applied in [14,15] and an extended state observer based control was applied to coaxial-rotor in [1,16].

Backstepping is a well known technique extensively used to control the nonlinear systems [1]. However, in the presence of model uncertainties and disturbances, this algorithm can not guarantee the stability of the closed loop system and the asymptotic convergence of the tracking error. Several methodologies can be combined with backstepping to attain desirable characteristics of a control law, such as robustness to external disturbances and actuation boundedness. The first one known as adaptive backstepping method [17], which is able to reject the disturbance, once its estimation is available or measured. The second is called robust backstepping method [4], which is able to reject the disturbance, without needing information on its evolution. In general, the backstepping control technique can not applicable directly to control the underactuated RUAVs system.

The backstepping methodology for helicopters is constructed by taking into account a dynamic extension of the thrust actuation, and hence, the control equations becomes complex and difficult to implement. A common characteristic unifies these controller: the

existence of a singularity in the control law for zero thrust. Typically, the singular condition is either ignored or the control laws are modified when near the singularity, but that leads to a loss of the stability properties and can endanger a vehicle unnecessarily.

In this paper, we propose an original control algorithm based on hierarchical backstepping control driven by a finite time disturbance observer to solve the path tracking control of a small coaxial rotor vehicle subject to unknown uncertainties and aerodynamic disturbances. The hierarchical flight control is proposed to separate the flight control problem into high level position control and a low level attitude control. The question of modeling the aerodynamic effects is over thrown by considering them as unknown perturbations that act on the system. To address the issues of uncertainties and external disturbances, a new disturbance observer is designed to online estimate the unknown uncertainties and disturbances and to actively compensate them in finite time. The finite-time sliding mode control [18–20] and observer disturbance [21–23] are used to construct the presented disturbance observer. The proposed FTDO offers superior faster convergence, better disturbance rejection capability and robustness against the model uncertainties and disturbances. The aim of control action is to enhance the performance of hierarchical control working in the outer control loop with the observer operating in the inner perturbation reconstruction-rejection loop. The performance for closed-loop system can be recovered if the total uncertainty/disturbance is timely compensated via disturbance observer.

The rest of paper is organized as follows. Section 2 presents the dynamical model of the coaxial-rotor UAV. Our proposed control algorithm based on hierarchical backstepping intergrated with a new disturbance observer for coaxial rotors is then presented and detailed in Section 3. Simulation results are then presented in Section 4 and conclusion is given in Section 5.

2. Vehicle dynamics modeling

We describe in this section the dynamical model of a small coaxial-rotor UAV which represent the behavior of the real system over time. This system is an underactuated mechanical system with six degrees of freedom, and only four degrees of freedom controlled with four control inputs, the thrust T_z produced by the two rotors and the control torque $\Gamma_a = (\tau_\phi, \tau_\theta, \tau_\psi)^T$ produced by both rotors and Swash-plate incidence angles. The main thrust is used to compensate the gravity force and to control the vertical movement [3]. The horizontal movements are controlled by directing the force vector in the appropriate direction (thrust vectoring control) through the cyclic swash plate. Control moments are used to control the aircraft body orientation which controls the rotorcraft horizontal movement.

Consider the coaxial-rotor as a solid body evolving in a 3D space and subject to the main thrust and three torques [1,24] as depicted in Fig. 1. Let $\mathcal{B} := (G, x_b, y_b, z_b)$ the body-fixed frame attached to the center of gravity of the aerial vehicle, where x_b is the longitudinal axis, y_b is the lateral axis and z_b is the vertical direction in hover conditions and $\mathcal{I} := (O, x_I, y_I, z_I)$ is the Earth frame. The generalized coordinates describing the rotorcraft position and orientation are $q = [\xi, \eta]^T$, where $\xi = (x, y, z)^T \in \mathbb{R}^3$ represents the translation coordinates relative to the inertial frame and $\eta = (\phi, \theta, \psi)^T \in \mathbb{R}^3$ are the classic yaw, pitch and roll Euler angles commonly used in aerodynamic applications. The rotation matrix $R_\eta = R_\psi R_\theta R_\phi \in \text{SO}(3)$ is the rotation matrix between Earth and body coordinate systems given by [25]¹

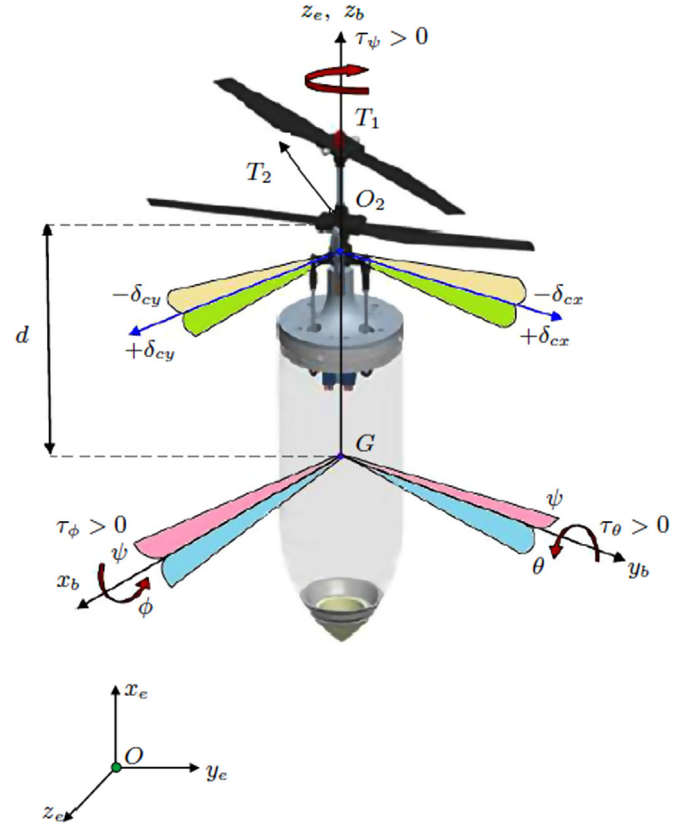


Fig. 1. Diagram showing the reference frame and forces of the coaxial-rotor UAV flight.

$$R_\eta = \begin{pmatrix} C_\theta C_\psi & S_\phi S_\theta C_\psi - C_\phi S_\psi & C_\phi S_\theta C_\psi + S_\phi S_\psi \\ C_\theta S_\psi & S_\phi S_\theta S_\psi + C_\phi C_\psi & C_\phi S_\theta S_\psi - S_\phi C_\psi \\ -S_\theta & S_\phi C_\theta & C_\phi C_\theta \end{pmatrix}$$

The dynamic model of the coaxial-rotor helicopter is expressed using Newton-Euler formulation [1,24]

$$\begin{cases} m\ddot{\xi} = R_\eta T - mgz_e + F_{ext} \\ \mathbb{J}\ddot{\eta} = \Gamma_a - C(\eta, \dot{\eta})\dot{\eta} + \Gamma_{ext} \end{cases} \quad (1)$$

where $\mathbb{J} = J\Psi(\eta) \in \mathbb{R}^{3 \times 3}$ is an auxiliary positive inertia matrix provided that $(\theta \neq k\pi/2)$, $m \in \mathbb{R}$ specifies the mass, $J \in \mathbb{R}^{3 \times 3}$ is the diagonal inertia matrix, g the acceleration due to a gravity and $C(\eta, \dot{\eta})\dot{\eta}$ is given by

$$C(\eta, \dot{\eta}) = \mathbb{J}\dot{\eta} - sk(\Psi\dot{\eta})\mathbb{J}\dot{\eta}$$

The attitude kinematic matrix $\Psi(\eta) \in \mathbb{R}^{3 \times 3}$ and the skew anti-symmetric matrix $sk(\beta)$ of β are defined, respectively, as [18]

$$\Psi(\eta) = \begin{pmatrix} 1 & 0 & -s_\theta \\ 0 & c_\phi & s_\phi c_\theta \\ 0 & -s_\phi & c_\phi c_\theta \end{pmatrix}; \quad sk(\beta) = \begin{pmatrix} 0 & -\beta_3 & \beta_2 \\ \beta_3 & 0 & -\beta_1 \\ -\beta_2 & \beta_1 & 0 \end{pmatrix}$$

2.1. Forces acting on the vehicle

The thrust vector generated by the two rotors is a function of the rotor angular speed and the cyclic tilt angles. The upper rotor has no swashplate, then, produces only the vertical thrust, whereas the lower rotor generates both a vertical thrust and lateral forces due to the swashplate incidence angles. Then, the total thrust vector T is defined [24]

¹ The abbreviations s_* and c_* denote $\sin(*)$ and $\cos(*)$, respectively.

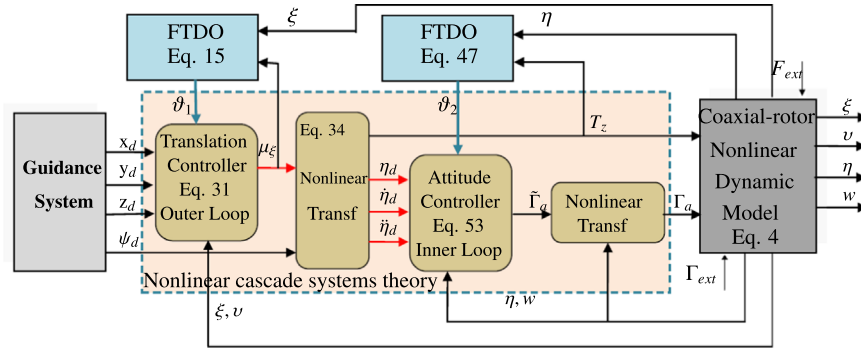


Fig. 2. Structure of the inner-outer-loop nonlinear controller.

$$T = \begin{pmatrix} T_x \\ T_y \\ T_z \end{pmatrix} = \begin{pmatrix} -\kappa_\beta \sin \delta_{cy} \cos \delta_{cx} \Omega_2^2 \\ -\kappa_\beta \sin \delta_{cx} \Omega_2^2 \\ \kappa_\alpha \Omega_1^2 + \kappa_\beta \cos \delta_{cx} \cos \delta_{cy} \Omega_2^2 \end{pmatrix} \quad (2)$$

where $\kappa_\beta > 0$ and $\kappa_\alpha > 0$ are the rotor aerodynamic coefficients, Ω_1 and Ω_2 the rotors rotation speeds and δ_{cx} and δ_{cy} the swashplate incidence angles.

In addition, there exist some others aerodynamic forces acting on the fuselage including the external wind gusts, internal couplings and unmodelled dynamics. These force disturbances directly affect the translational dynamics and result in tracking error. Let us denote by F_{ext} all the aerodynamic efforts viewed as an unknown disturbance acting on the system in the inertial frame.

2.2. Moments acting on the vehicle

In rotation dynamics, the thrust vector T generates the torque vector due to the separation between the centre of the mass G and the rotor hubs. According to [24], the total torque vector Γ_a applied on the airframe is given by

$$\Gamma_a = \begin{pmatrix} \tau_\phi \\ \tau_\theta \\ \tau_\psi \end{pmatrix} = \begin{pmatrix} -d\kappa_\beta \sin \delta_{cx} \Omega_2^2 \\ d\kappa_\beta \sin \delta_{cy} \cos \delta_{cx} \Omega_2^2 \\ \gamma_1 \Omega_1^2 - \gamma_2 \Omega_2^2 \end{pmatrix} \quad (3)$$

d being the distance between the centre of gravity and the lower rotor center of rotation and $\gamma_1 > 0$ and $\gamma_2 > 0$ are the yaw aerodynamic coefficients.

Additionally, the moment Γ_{ext} is given by $\Gamma_{ext} = l_{ext} \wedge F_{ext}$, it results from the aerodynamic effects with a lever arm l_{ext} .

According to (3), the force vector in (2) can be rewritten as follows

$$T = (0, 0, T_z)^T + \Sigma \Gamma_a \quad \text{with} \quad \Sigma = \frac{1}{d} sk(z_e)$$

The matrix Σ displays the coupling between the mechanism force and moment generation which produces small parasitic forces named small body forces.

2.3. The complete model

By merging (2), (3) in a Newton Euler model (1), The equations of motion for a coaxial helicopter become [1]

$$\begin{pmatrix} \ddot{\xi} \\ m\dot{v} \\ \dot{\eta} \\ J\dot{w} \end{pmatrix} = \begin{pmatrix} v \\ T_z R_\eta z_e - mgz_e + R_\eta \Sigma \Gamma_a + F_{ext} \\ w \\ \Gamma_a - C(\eta, w)w + \Gamma_{ext} \end{pmatrix} \quad (4)$$

$v = (v_x, v_y, v_z)^T$ and $w = (w_\phi, w_\theta, w_\psi)^T$ are the linear and angular velocities in $\mathcal{I}$, respectively.

Remark 1. From a practical point of view, the model may be simplified knowing that the lateral tilting (δ_{cx}, δ_{cy}) are imposed to be small. Therefore, we can consider that ($\cos \delta_{ci} \approx 1$) and ($\sin \delta_{ci} \approx \delta_{ci}$). Consequently, from the expression (2) and (3), the two lateral forces (T_x, T_y) due to small cyclic swashplate incidence angles (δ_{cx}, δ_{cy}) are small with respect to the vertical thrust and consequently ($\Sigma \Gamma_a \approx 0$).

It is important to note that the change of variables in (2) and (3) under the remark.1 defines a diffeomorphism. In other words, the original control inputs ($\Omega_1, \Omega_2, \delta_{cx}, \delta_{cy}$) can be recovered from (T_z, Γ_a) by applying the inverse transformation, which are given by

$$\Omega_1^2 = \frac{\gamma_2 T_z - \kappa_\beta \tau_\psi}{\kappa_\alpha \gamma_2 + \kappa_\beta \gamma_1}, \quad \Omega_2^2 = \frac{\gamma_1 T_z - \kappa_\alpha \tau_\psi}{\kappa_\beta \gamma_1 + \kappa_\alpha \gamma_2}, \quad \delta_{cx} = -\frac{\tau_\phi}{d\kappa_\beta \Omega_2^2},$$

$$\delta_{cy} = \frac{\tau_\theta}{d\kappa_\beta \Omega_2^2}$$

3. Control design

In this section, a new flight control approach is proposed by combining the hierarchical backstepping with a disturbance observer (FTDO) to solve the path tracking problem for a small coaxial helicopter in the presence of unknown aerodynamic efforts. The resulting nonlinear controller is thus, simple, robust and easy to implement. The aim of control action in the proposed methodology is to enhance the performance of conventional controller working in the outer control loop with the disturbance observer operating in the inner perturbation reconstruction-rejection loop. The basic idea for designing a hierarchical flight controller is to split the controller into an inner-loop that controls the moments acting on the aircraft, and an outer-loop that controls the thrust force acting on the aircraft (Fig. 2). In fact, the thrust vector is effectively oriented in the desired direction by controlling changes to the UAV attitude using the inner-loop.

The following lemma is used for the stability analysis [26].

Lemma 1. [26] Consider the continuous system

$$\dot{x} = f(x), \quad f(0) = 0, \quad x \in \mathbb{R}^n \quad (5)$$

Suppose there exists a continuous positive definite function $V: \mathbb{R}^n \rightarrow \mathbb{R}$, a real number $\gamma > 0$ and $\alpha \in (0, 1)$ and an open neighborhood $U_0 \subseteq \mathbb{R}^n$ of the origin such that the following inequality is satisfied

$$\dot{V}(x) + \gamma V^\alpha(x) \leq 0, \quad x \in \hat{U} \setminus \{0\}.$$

Then, the origin of the system (5) is finite time stable. If $U_0 = \mathbb{R}^n$, then the origin is globally finite time stable. The settling time t_R satisfies

$$t_R \leq \frac{1}{\gamma(1-\alpha)} V^{1-\alpha}$$

3.1. Control strategy

The hierarchical backstepping control design procedure integrated with disturbance observer with finite time convergence for the coaxial rotor is stated in the following step:

Step 1. Let ξ_d be the reference trajectory of the translational position ξ . Introducing the error between the actual position and the desired one

$$\delta_1 = \xi - \xi_d \quad (6)$$

Consider the Lyapunov function candidate $\mathcal{L}_1 = \frac{1}{2} \delta_1^T \delta_1$. Differentiating $\mathcal{L}_1$ with respect to time yields

$$\dot{\mathcal{L}}_1 = \delta_1^T \dot{\delta}_1 = \delta_1^T (\rho_1 - \dot{\xi}_d) + \delta_1^T (v - \rho_1) \quad (7)$$

The control law must ensure the convergence of δ_1 towards zero, the stabilization of δ_1 can be ensured by introducing the first internal control input ρ_1 as

$$\rho_1 = \dot{\xi}_d - K_1 \delta_1 \quad (8)$$

where $K_1 \in \mathbb{R}_{+}^{3 \times 3}$. The second term of (7) $\delta_2 = v - \rho_1$ represents the difference between the real velocity v and a internal control which, if $\delta_2 = 0$ would lead to the exponential convergence of δ_1 , we focus only on the first term of (7), the second one will be eliminated in the next step, then substituting (8) into (7) yields

$$\dot{\mathcal{L}}_1 = -K_1 \|\delta_1\|^2 \leq 0 \quad (9)$$

Step 2. The process of backstepping continues by considering

$$\delta_2 = v - \rho_1 \quad (10)$$

as a new auxiliary error signal. Differentiating (10) with respect to time, we get

$$\dot{\delta}_2 = \frac{T_z}{m} R_{\eta} z_e - g z_e - \Psi_1 + \ddot{F}_{ext} \quad (11)$$

where $\ddot{F}_{ext} = \frac{\dot{F}_{ext}}{m}$ and

$$\Psi_1 = K_1 v - (\dot{\xi}_d + K_1 \dot{\xi}_d)$$

The term $\ddot{F}_{ext}$ present the external disturbance acting the translational dynamic, including the drag forces and the rotor air slipstream and unknown perturbation acting on the system (wind gusts). This term assume to be unknown but bounded. To stabilize the system, the convergence of δ_2 must then be ensured. In the expression (11), the unknown vector $\ddot{F}_{ext}$ appears via the translation dynamics. The term $\ddot{F}_{ext}$ will be compensated using a disturbance observer with finite time convergence (FTDO) to ensure the robustness of the control strategy.

3.2. Design of Disturbance Observer

In this section, a disturbance observer with finite time convergence (FTDO) is designed to estimate the unknown total disturbance $\ddot{F}_{ext}$. The disturbance observer dynamic for the system (11) is given by

$$\dot{\hat{\delta}}_2 = \frac{T_z}{m} R_{\eta} z_e - g z_e - \Psi_1 + \vartheta_1 \quad (12)$$

The observation error can be written as

$$\tilde{\delta}_2 = \delta_2 - \hat{\delta}_2 \quad (13)$$

From (11) and (12), the observer error dynamics can be written as

$$\dot{\tilde{\delta}}_2 = \ddot{F}_{ext} - \vartheta_1 \quad (14)$$

The injection terms ϑ_1 is given in a finite-time format as

$$\vartheta_1 = \lambda_0 \text{sig}^{\sigma_1}(\tilde{\delta}_2) + \lambda_1 \int \text{sig}^{\sigma_2}(\tilde{\delta}_2(\tau)) d\tau \quad (15)$$

where $\text{sig}^{\sigma}(\tilde{\delta}_2) = |\tilde{\delta}_2|^{\sigma} \text{sign}(\tilde{\delta}_2)$, $\sigma_1 \in]1/2, 1[$ and $\sigma_2 = 2\sigma_1 - 1$. Substituting (15) into (14), one get

$$\dot{\tilde{\delta}}_2 = -\lambda_0 \text{sig}^{\sigma_1}(\tilde{\delta}_2) - \lambda_1 \int \text{sig}^{\sigma_2}(\tilde{\delta}_2(\tau)) d\tau + \ddot{F}_{ext} \quad (16)$$

Now, consider the following change of variable

$$z = [z_1 \ z_2]^T = [\tilde{\delta}_2 \ \varpi]^T$$

where $\varpi = -\lambda_1 \int \text{sig}^{\sigma_2}(\tilde{\delta}_2(\tau)) d\tau + \ddot{F}_{ext}$. Then, the system (16) becomes

$$\begin{aligned} \dot{z}_1 &= -\lambda_0 |z_1|^{\sigma_1} \text{sign}(z_1) + z_2 \\ \dot{z}_2 &= -\lambda_1 |z_1|^{\sigma_2} \text{sign}(z_1) + q_1(t) \end{aligned} \quad (17)$$

where $q_1(t)$ is the time derivative of $\ddot{F}_{ext}$ which is assumed bounded by $\|q_1(t)\| \leq h_{1 \max}$, with $h_{1 \max}$ is a known constant.

It can be noted that if $z_1, z_2 \rightarrow 0$ in finite time, then $\dot{z}_1 \rightarrow 0$ and consequently $\vartheta_1 = \ddot{F}_{ext}$ in finite time.

Consider now the first auxiliary Lyapunov candidate function

$$\mathcal{L}_{ob1}(z) = \zeta_1^T P_1 \zeta_1 \quad (18)$$

The vector ζ_1 and the symmetric positive definite matrix $P_1 \in \mathbb{R}^{2 \times 2}$ are selected respectively as

$$\zeta_1 = \begin{pmatrix} \text{sig}^{\sigma_1}(z_1) \\ z_2 \end{pmatrix}, \quad P_1 = \frac{1}{2\sigma_1} \begin{pmatrix} 2\lambda_1 + \sigma_1 \lambda_0^2 & -\sigma_1 \lambda_0 \\ -\sigma_1 \lambda_0 & 2\sigma_1 \end{pmatrix}$$

Note that the function (18) is continuous and differentiable everywhere except on the set $z = \{(z_1, z_2) \in \mathbb{R}^2 | z_1 = 0\}$, it is positive definite and radically unbounded if $\lambda_1 > 0$, hence, it follows that $\mathcal{L}_{ob1}(z)$ satisfies

$$\lambda_{\min}(P_1) \|\zeta_1\|^2 \leq \mathcal{L}_{ob1}(z) \leq \lambda_{\max}(P_1) \|\zeta_1\|^2 \quad (19)$$

where

$$\|\zeta_1\|^2 = |z_1|^{2\sigma_1} + z_2^2 \quad (20)$$

The time derivative of the Lyapunov function candidate (18) verify

$$\begin{aligned} \dot{\mathcal{L}}_{ob1}(z) &\leq -|z_1|^{\sigma_1-1} \zeta_1^T Q_1 \zeta_1 + q_1 q_1^T \zeta_1 \leq -|z_1|^{\sigma_1-1} \lambda_{\min}(Q_1) \\ &\quad \|\zeta\|^2 + h_{1 \max} \|q_1\| \|\zeta_1\| \end{aligned} \quad (21)$$

where

$$Q_1 = \lambda_0 \begin{pmatrix} \lambda_1 + \sigma_1 \lambda_0^2 & -\sigma_1 \lambda_0 \\ -\sigma_1 \lambda_0 & \sigma_1 \end{pmatrix}; \quad q_1 = \begin{pmatrix} -\lambda_0 \\ 2 \end{pmatrix}$$

From (20), we can get

$$|z_1|^{\sigma_1-1} \geq \|\zeta_1\|^{\frac{\sigma_1-1}{\sigma_1}} \quad (22)$$

From (21) and (22), we get

$$\dot{\mathcal{L}}_{ob1}(z) \leq -\left(\lambda_{\min}(Q_1) - \frac{h_{1 \max} \|q_1\|}{\|\zeta_1\|^{\frac{\sigma_2}{\sigma_1}}} \right) \|\zeta_1\|^{\frac{\sigma_1+\sigma_2}{\sigma_1}} \quad (23)$$

Let

$$\nu_1 = \lambda_{\min}(Q_1) - \frac{h_{1 \max} \|q_1\|}{\|\zeta_1\|^{\frac{\sigma_2}{\sigma_1}}} \quad (24)$$

with

$$\| \zeta_1 \| > \left(\frac{h_{1 \max} \| q_1 \|}{\lambda_{\min}(Q_1)} \right)^{\frac{\sigma_1}{\sigma_2}} \quad (25)$$

to ensure $\nu_1 > 0$. From (19) and (24), it can be easily shown that the expression (23) will be bounded as follows

$$\dot{\mathcal{L}}_{ob1}(z) \leq -\gamma_1 \mathcal{L}_{ob1}^\alpha(z) \quad (26)$$

with $\alpha = \frac{\sigma_1 + \sigma_2}{2\sigma_1}$, then $\alpha \in (0, 1)$, and the positive constant $\gamma_1 = \left[\nu_1 \frac{\lambda_{\min}(P_1)^{2\sigma_1}}{\sqrt{\lambda_{\min}(P_1)}} \right] > 0$, which show that $\mathcal{L}_{ob1}(z)$ is a strong Lyapunov function. According to lemma 1, the finite time stability can be guaranteed, and hence, all trajectories of the system (17) converge in finite time to the region

$$\| \zeta_1 \| \leq \left(\frac{h_{1 \max} \| q_1 \|}{\lambda_{\min}(Q_1)} \right)^{\frac{\sigma_1}{\sigma_2}} \quad (27)$$

In fact, we can choose λ_0 and λ_1 such that $(h_{1 \max} \| q_1 \| / \lambda_{\min}(Q_1) < 1)$. Because (σ_1/σ_2) is sufficiently large, $\| \zeta_1 \|$ is sufficiently small in finite time. Therefore, we can select λ_0 and λ_1 such that

$$\lambda_0 > 0, \quad \lambda_1 > h_{1 \max}^2 \| q_1 \|^2 / \sigma_1 \lambda_0^2$$

Remark 2. Observing that if we take the value of the fraction power $\sigma_1 = 1/2$ a strong robustness property is also ensured and the trajectories of the system (15) converge in finite time to the origin ($z_1 = 0$) and ($\dot{z}_1 = 0$).

According to (13)–(27), The equation (11) will be

$$\delta_2 = \frac{T_z}{m} R_\eta z_e - g z_e - \Psi_1 + \vartheta_1 \quad (28)$$

Consider now the augmented Lyapunov function candidate $\mathcal{L}_2 = \mathcal{L}_1 + \frac{1}{2} \delta_2^T \delta_2$. Differentiating $\mathcal{L}_2$ with respect to time yields

$$\begin{aligned} \dot{\mathcal{L}}_2 = & -K_1 \| \delta_1 \|^2 + \delta_1^T \delta_2 + \delta_2^T \left[\mu_\xi - \Psi_1 + \vartheta_1 \right] \\ & + \delta_2^T \left[\frac{T_z}{m} R_\eta z_e - g z_e - \mu_\xi \right] \end{aligned} \quad (29)$$

where

$$\mu_\xi = \frac{T_z^d}{m} R_\eta^d z_e - g z_e \quad (30)$$

represent the translational controller used to extract the desired thrust T_z^d and the attitude matrix R_η^d . T_z is considered as the desired value T_z^d (i.e. $T_z^d = T_z$). The desired attitude information can enable a rotorcraft to track the desired trajectory.

The stabilisation of δ_2 can be ensured by introducing the internal control input μ_ξ :

$$\mu_\xi = \Psi_1 - \vartheta_1 - \delta_1 - K_2 \delta_2 \quad (31)$$

where $K_2 \in \mathbb{R}_{*+}^{2 \times 2}$. Substituting (31) into (29), we get

$$\dot{\mathcal{L}}_2 = -K_1 \| \delta_1 \|^2 - K_2 \| \delta_2 \|^2 + \delta_2^T \delta_e \quad (32)$$

The internal control input μ_ξ includes the desired Euler angles η_d . Introduce the next error variable as follows

$$\delta_e = \mu_\xi - \frac{T_z}{m} R_\eta z_e + g z_e \quad (33)$$

The application of the conventional backstepping control to the rotorcraft UAV would require the definition of a third tracking error δ_e . But This would also lead to a more complex control equations. Based on the hierarchical backstepping, the coaxial-

rotor model can be separated into two connected subsystems by decoupling the translational dynamics and rotational dynamics. The rotation dynamics do not depend on translation components, while the translational dynamics depend on angles via the rotation matrix R_η^d .

According to (30), we can calculate the desired force vector magnitude and the desired R_η^d as follows [1]:

$$\begin{cases} T_z = m \sqrt{\mu_x^2 + \mu_y^2 + (\mu_z + g)^2} \\ \phi_d = \sin^{-1} \left[\frac{m}{T_z} (\mu_x \sin \psi_d - \mu_y \cos \psi_d) \right] \\ \theta_d = \sin^{-1} \left[\frac{m}{T_z \cos \psi_d} (\mu_x \cos \psi_d + \mu_y \sin \psi_d) \right] \end{cases} \quad (34)$$

Step 3. A new error variable is then defined as

$$\delta_3 = \eta - \eta_d \quad (35)$$

Let us take $\eta = \eta_d + \delta_3$, then the vector $\frac{T_z}{m} R_\eta z_e$ can be rewritten as

$$\frac{T_z}{m} R_\eta z_e = \frac{T_z}{m} R_\eta^d z_e + \frac{1}{m} h(\eta_d, \delta_3) \quad (36)$$

The vector $h(\eta_d, \delta_3)$ represents the interconnection term between the translation and rotation dynamics and can be found in [1]. Substituting (36) into (33), one get

$$\delta_e = \frac{1}{m} h(\eta_d, \delta_3) \quad (37)$$

According to [1] one can easily show that the interconnection term $h(\eta_d, \delta_3)$ is bounded and if $\delta_3 = 0$, then $h(\eta_d, \delta_3) = 0$ and consequently $\delta_e = 0$.

Consider the third Lyapunov function candidate $\mathcal{L}_3 = \frac{1}{2} \delta_3^T \delta_3$. Differentiating $\mathcal{L}_3$ with respect to time yields

$$\dot{\mathcal{L}}_3 = \delta_3^T \dot{\delta}_3 = \delta_3^T (\rho_3 - \dot{\eta}_d) + \delta_3^T (\omega - \rho_3) \quad (38)$$

where $\dot{\eta}_d$ denotes a desired value for the angular velocity. The stabilization of δ_3 can be ensured by introducing the third internal control input ρ_3 as:

$$\rho_3 = \dot{\eta}_d - K_3 \delta_3 \quad (39)$$

where $K_3 \in \mathbb{R}_{*+}^{3 \times 3}$, we focus only on the first term of (38), the second one will be eliminated in the next step, then substituting (39) into (38) yields

$$\dot{\mathcal{L}}_3 = -K_3 \| \delta_3 \|^2 \leq 0 \quad (40)$$

Step 4. The process of backstepping continues by considering the second term of (38)

$$\delta_4 = \omega - \rho_3 \quad (41)$$

as a new auxiliary error signal. Differentiating (41) with respect to time, we get

$$\dot{\delta}_4 = \mathbb{J}^{-1} [\Gamma_a - C(\eta, w)w + \Gamma_{ext}] \quad (42)$$

By considering the following change of variable

$$\Gamma_a = \mathbb{J} \bar{\Gamma}_a, \quad \Gamma_{ext} = \mathbb{J} \bar{\Gamma}_{ext} \quad \text{and} \quad \Psi_2 = \mathbb{J}^{-1} C(\eta, w)w$$

The expression (42) will be

$$\dot{\delta}_4 = \bar{\Gamma}_a - \Psi_2 + \bar{\Gamma}_{ext} \quad (43)$$

The term $\bar{f}_{ext}$ present the external disturbance acting on the rotational dynamics, which assumed to be unknown but bounded. To stabilize the system, the convergence of δ_4 must then be ensured. In the expression (43), the unknown vector $\bar{f}_{ext}$ appears via the rotational dynamics. This term will be compensated using a second disturbance observer with finite time convergence (FTDO) to ensure the robustness of the control strategy.

In order to design the second disturbance observer with finite time convergence to estimate the bounded disturbance $\bar{f}_{ext}$. The disturbance observer dynamic for the system (43) is given by

$$\dot{\delta}_4 = \bar{f}_a - \Psi_2 + \vartheta_2 \quad (44)$$

The observation error can be written as

$$\tilde{\delta}_4 = \delta_4 - \hat{\delta}_4 \quad (45)$$

From (43) and (44), the observer error dynamics becomes

$$\dot{\tilde{\delta}}_4 = \bar{f}_{ext} - \vartheta_2 \quad (46)$$

The injection terms ϑ_2 is given in a finite-time format as

$$\vartheta_2 = \lambda_2 \text{sig}^{\sigma_1}(\tilde{\delta}_4) + \lambda_3 \int \text{sig}^{\sigma_2}(\tilde{\delta}_4(\tau)) d\tau \quad (47)$$

Substituting (47) into (46), one get

$$\dot{\tilde{\delta}}_4 = -\lambda_2 \text{sig}^{\sigma_1}(\tilde{\delta}_4) - \lambda_3 \int \text{sig}^{\sigma_2}(\tilde{\delta}_4(\tau)) d\tau + \bar{f}_{ext} \quad (48)$$

Now, consider the following change of variable

$$\bar{z} = [\bar{z}_1 \ \bar{z}_2]^T = [\tilde{\delta}_4 \ \bar{\omega}]^T \quad (49)$$

where $\bar{\omega} = -\lambda_3 \int \text{sig}^{\sigma_2}(\tilde{\delta}_4(\tau)) d\tau + \bar{f}_{ext}$. Then, the system (48) become

$$\begin{aligned} \dot{\bar{z}}_1 &= -\lambda_2 |\bar{z}_1|^{\sigma_1} \text{sign}(\bar{z}_1) + \bar{z}_2 \\ \dot{\bar{z}}_2 &= -\lambda_3 |\bar{z}_1|^{\sigma_2} \text{sign}(\bar{z}_1) + q_2(t) \end{aligned} \quad (50)$$

where $q_2(t)$ is the time derivative of $\bar{f}_{ext}$ which is assumed bounded by $\|q_2(t)\| \leq h_{2\max}$, where $h_{2\max}$ is a known constant.

It can be noted that if $\bar{z}_1, \bar{z}_2 \rightarrow 0$ in finite time, then $\tilde{\delta}_4, \dot{\tilde{\delta}}_4 \rightarrow 0$ and consequently $\vartheta_2 = \bar{f}_{ext}$ in finite time.

According to (44)–(50), The equation (43) will be

$$\delta_4 = \bar{f}_a - \Psi_2 + \vartheta_2 \quad (51)$$

Consider now the augmented Lyapunov function candidate $\mathcal{L}_4 = \mathcal{L}_3 + \frac{1}{2} \delta_4^T \delta_4$. Differentiating $\mathcal{L}_4$ with respect to time yields

$$\dot{\delta}_4 = -K_3 \|\delta_3\|^2 + \delta_3^T \delta_4 + \delta_4^T [\bar{f}_a - \Psi_2 + \vartheta_2] \quad (52)$$

The stabilisation of δ_4 can be ensured by introducing the control input

$$\bar{f}_a = \Psi_2 - \delta_3 - \vartheta_2 - K_4 \delta_4 \quad (53)$$

where $K_3 \in \mathbb{R}_{+*}^{3 \times 3}$. Substituting (53) into (52), we get

$$\dot{\delta}_4 = -K_3 \|\delta_3\|^2 - K_4 \|\delta_4\|^2 \leq 0 \quad (54)$$

3.3. Closed Loop Dynamics

Applying the control laws (8) and (31) to the system (4), the closed-loop translational dynamics may be written in error coordinates as:

$$\begin{aligned} \dot{\delta}_1 &= -K_1 \delta_1 + \delta_2 \\ \dot{\delta}_2 &= -K_2 \delta_2 - \delta_1 + \delta_e \end{aligned} \quad (55)$$

with the corresponding Lyapunov function

$$\mathcal{L}_2 = \frac{1}{2} \delta_1^T \delta_1 + \frac{1}{2} \delta_2^T \delta_2 \quad (56)$$

and its derivative

$$\dot{\mathcal{L}}_2 = -K_1 \|\delta_1\|^2 - K_2 \|\delta_2\|^2 + \delta_2^T \delta_e \quad (57)$$

The closed-loop orientation dynamics, it may be written in error coordinates as:

$$\begin{aligned} \dot{\delta}_3 &= -K_3 \delta_3 + \delta_4 \\ \dot{\delta}_4 &= -K_4 \delta_4 - \delta_3 \end{aligned} \quad (58)$$

with the corresponding Lyapunov function

$$\mathcal{L}_4 = \frac{1}{2} \delta_3^T \delta_3 + \frac{1}{2} \delta_4^T \delta_4 \quad (59)$$

and its derivative

$$\dot{\mathcal{L}}_4 = -K_3 \|\delta_3\|^2 - K_4 \|\delta_4\|^2 \quad (60)$$

Noting that δ_e vanishes as $\delta_3 \rightarrow 0$ (from Eq (37)), the controlled system has the structure of an upper subsystem describing the translational dynamics in cascade with a lower subsystem describing the orientation dynamics.

From Eqs. (58)–(60) it is clear that the subsystem describing the orientation dynamics is globally asymptotically stable. It is indeed a straightforward application of Lyapunov stability theory and LaSalle's invariance principle. From Eqs. (55)–(57), and for the same reasons, the translational subsystem is also globally asymptotically, with the exception of the term δ_e . However, this term will vanish as δ_e converges to zero.

Remark 3. The final control inputs ($T_z, \bar{f}_a$) are given by

$$\begin{aligned} T_z &= m \|\mu_\xi + gZ_e\| = m \|\Psi_1 - \vartheta_1 - \delta_1 - K_2 \delta_2 + gZ_e\| \\ \bar{f}_a &= \mathbb{J}[\Psi_2 - \delta_3 - \vartheta_2 - K_4 \delta_4] \end{aligned} \quad (61)$$

4. Simulation results and discussions

In this section, the effectiveness and the robustness of the proposed nonlinear hierarchical backstepping integrated with a new disturbance observer are evaluated through various simulation. For convenience, the numerical values of the main physical parameters of the coaxial-rotor unmanned helicopter are provided in Table 1.

The main controller parameters are the same in the simulations, which are summarized in Table 2.

The simulation initial conditions for the translationam system are $\xi(0) = [0, 0, 0.5]^T$ m and for rotational system are: $\eta(0) = [0, 0, 0.3]^T$ rad. For, the simulation process, the initial values for the velocity and angular velocity are set to 0. The model uncertainty and aerodynamic forces and moments disturbances

Table 1
Parameters of the CRUAV [1].

Parameter	Value	Unit
g	9.81	m/s ²
m	0.41	kg
d	0.0676	m
J_x	1.383×10^{-3}	kg m ²
J_y	1.383×10^{-3}	kg m ²
J_z	2.72×10^{-4}	kg m ²
κ_α	3.6835×10^{-5}	N/rad ² s ²
κ_β	3.7760×10^{-5}	N/rad ² s ²
γ_1	1.4765×10^{-6}	N.m/rad ² s ²
γ_2	1.3266×10^{-6}	N.m/rad ² s ²

Table 2
Control parameters.

Control Parameter	Value
$K_i, i=\{1,2,3,4\}$	[1,1,1]
$\lambda_0=\lambda_2$	[1,1,1]
$\lambda_1=\lambda_3$	[0.6,0.6,0.6]
σ_1	0.6

acting on the system are listed below:

- for the uncertainty $\Delta J = 0.5J$ kg. m².
- For the aerodynamic forces occurring at 10, 30 and 40 sec, respectively:

$$\bar{F}_{ext} = \begin{bmatrix} \sin(0.1t) \\ \sin(0.1t) \\ \sin(0.1t) \end{bmatrix} \text{ N,}$$

- For the aerodynamic moments occurring at 10 sec:

$$\Gamma_{ext} = \begin{bmatrix} 0.3 \sin(0.1t) \\ 0.3 \sin(0.1t) \\ 0.5 \sin(0.1t) \end{bmatrix} \text{ N. m}$$

4.1. Case a: Flight with aerodynamic force and moment disturbances (without disturbance observer)

This part involves the position and the attitude tracking control of the coaxial system in the presence of inertia uncertainty, aerodynamic forces and moments disturbances by using the hierarchical backstepping without a disturbance observer. Fig. 3 shows the evolution of the vehicle's position and orientation during its flight. The explanation of these figures are separated into two cases. The forces disturbances occurring in 10, 30 and 40 sec respectively, and the moments disturbances occurring at 10 Sec. It can be seen that the system trajectories track the reference trajectories before the injection of forces and moment disturbances, and consequently the proposed controller is able to drive all these state variables back to the new reference position, but when the aerodynamic forces and

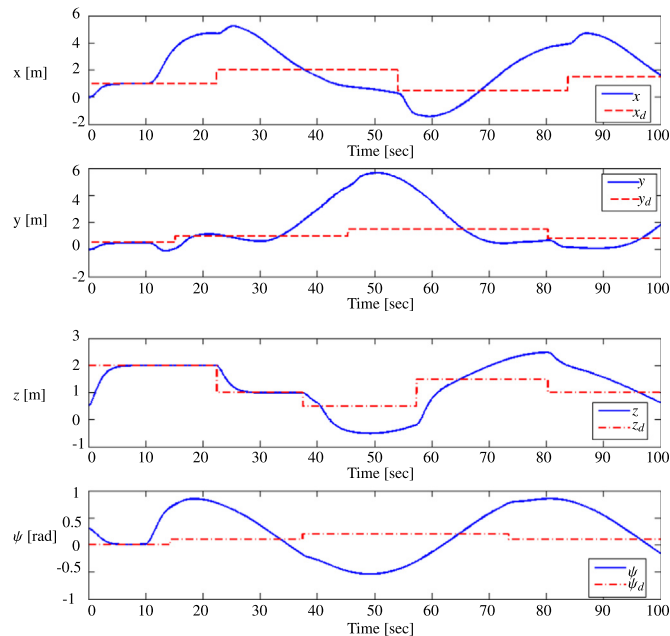


Fig. 3. Case a: Position outputs.

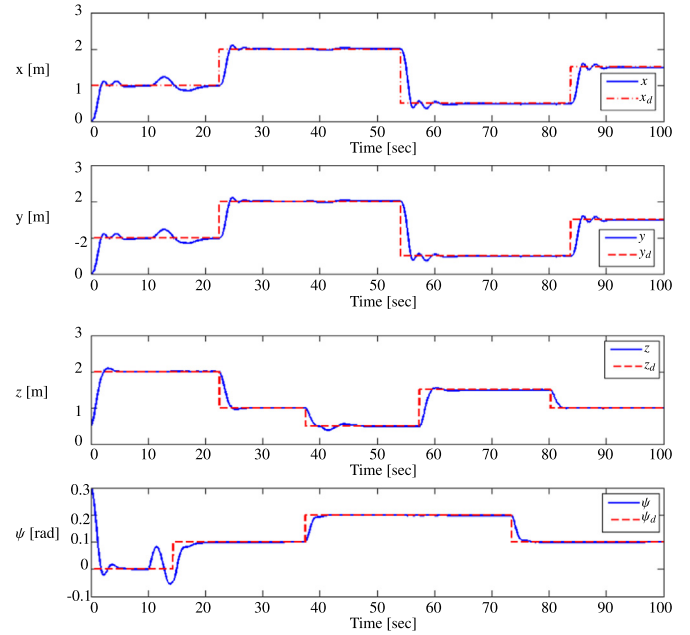


Fig. 4. Case b: Position outputs.

moments disturbances are introduced, the system cannot track the reference trajectories and the system becomes unstable, and consequently, the assigned navigational task cannot be achieved for the reference trajectories. From these figures, we can see that the hierarchical controller cannot ensure the stability of the system in the presence of external disturbances.

4.2. Case b: Flight with aerodynamic force and moment disturbances (with disturbance observer)

This part involves the position and attitude tracking control of the coaxial system in the presence of inertia uncertainty, aerodynamic forces and moments disturbances using the hierarchical backstepping connected with the disturbance observer. Fig. 4 shows the absolute position of the coaxial-rotor during its flight, when the aerodynamic forces and moments disturbances are introduced, the assigned navigational task are successfully achieved and the reference trajectories are tracked with high accuracy. From these figures, we can see a well good tracking of the desired trajectories, the disturbance observer is able to reject them showing the robustness of the proposed control approach and the stability of the closed loop dynamics is guaranteed. Fig. 5 shows the desired trajectory (ϕ_d, θ_d) generated by the translational controller in Eq. (34), and there are considered as the reference trajectories for the attitude system. These trajectories help the system to follow the reference trajectories generated by an operator or a guidance system. Finally, the actuator force commands, displayed on Fig. 6 are continuous as desired and could easily be applied to a real-life model.

4.3. Case c: Flight with sensing noise

The same numerical simulation as before was processed, by considering the inertia uncertainty, aerodynamic forces and moments disturbances, but considering now a white gaussian noise with 0.01 variance to position velocity sensors signals and a white gaussian noise with 0.0001 variance to the attitude velocity sensors signals. Thus, realistic uncertain signals are used for control purpose. Despite the sensing noise and the control inputs noise, the Figs. 7 and 8 are similar to the Figs 4 and 5, presented in the previous section. From these figures, the designed controller,

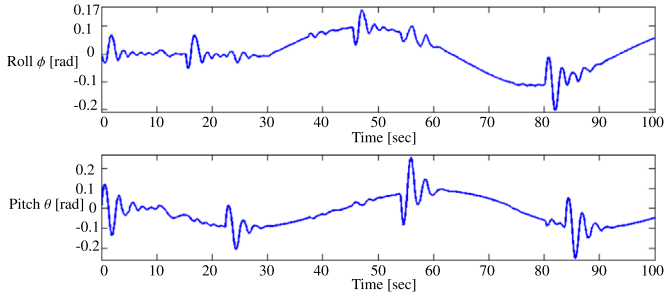


Fig. 5. Case b: Roll & Pitch trajectories.

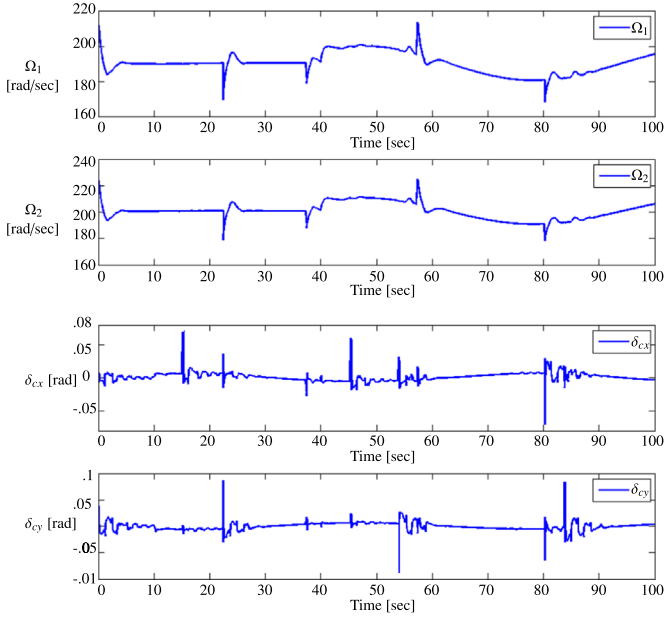
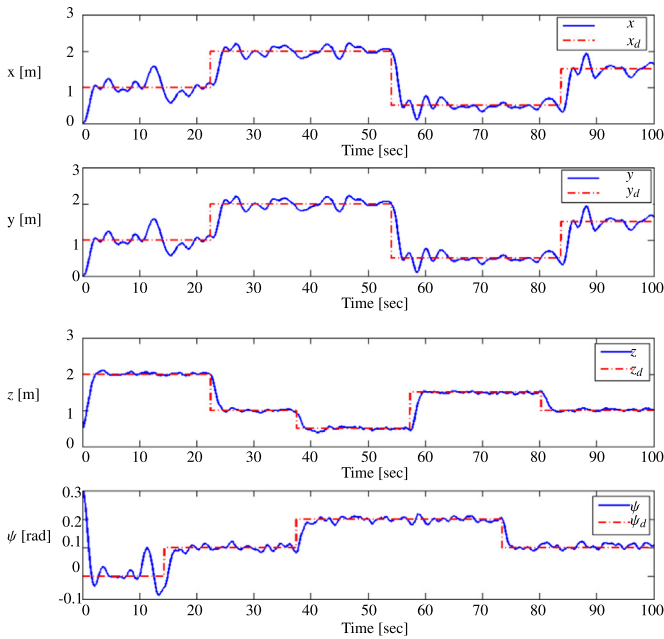
Fig. 6. Case b: Control signals Ω_1 , Ω_2 , δ_{cx} and δ_{cy} .

Fig. 7. Case c: Position outputs.

succeeds in reaching the desired position. Nevertheless, depending on the dynamics of the engines, the erratic property of the control signals shown in Fig. 9, which is not a problem here, may

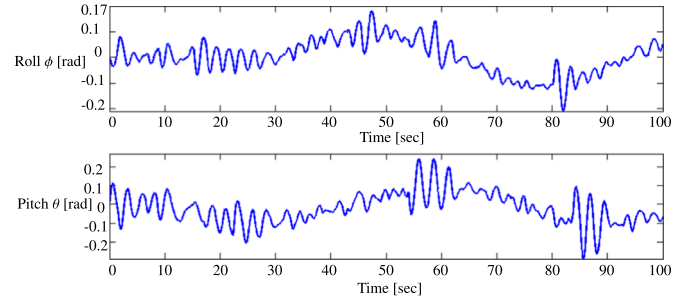
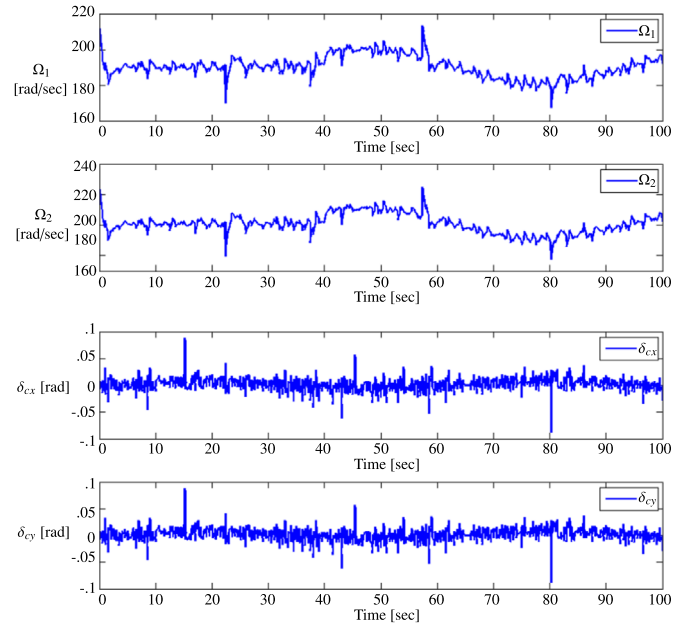


Fig. 8. Case c: Roll & Pitch trajectories.

Fig. 9. Case c: Control signals Ω_1 , Ω_2 , δ_{cx} and δ_{cy} .

be of paramount importance while moving from simulations towards the complex real world scenario of the future experimental validation.

4.4. Case d: Path tracking

The same numerical simulation as the previous cases was processed now by considering an aggressive flight. Fig. 10a shows the evolution of the vehicle's position and orientation during its flight (similar to case a). It can be seen that the system trajectories can not track the reference trajectories and the system becomes unstable, and consequently, the assigned navigational task cannot be achieved for the reference trajectories. Fig. 10b shows the evolution of the vehicle's position and orientation similarly to the case b. It can be seen that the system trajectories track the reference trajectories before the injection of forces and moment disturbances, and consequently the proposed controller is able to drive all these state variables back to the new reference position and Finally the Fig. 10c shows the evolution of the vehicle's position and orientation similarly to the case c. It can be seen that even though the commanded position $\{x_d(t), y_d(t), z_d(t), \psi_d(t)\}$ were changed in every moment, the proposed controller is able to drive all these state variables back to the new reference position. In addition, the proposed control strategy is able to make the vehicle follow the reference trajectory. From this figure, it is clear that the designed controller succeeds in reaching the desired position.

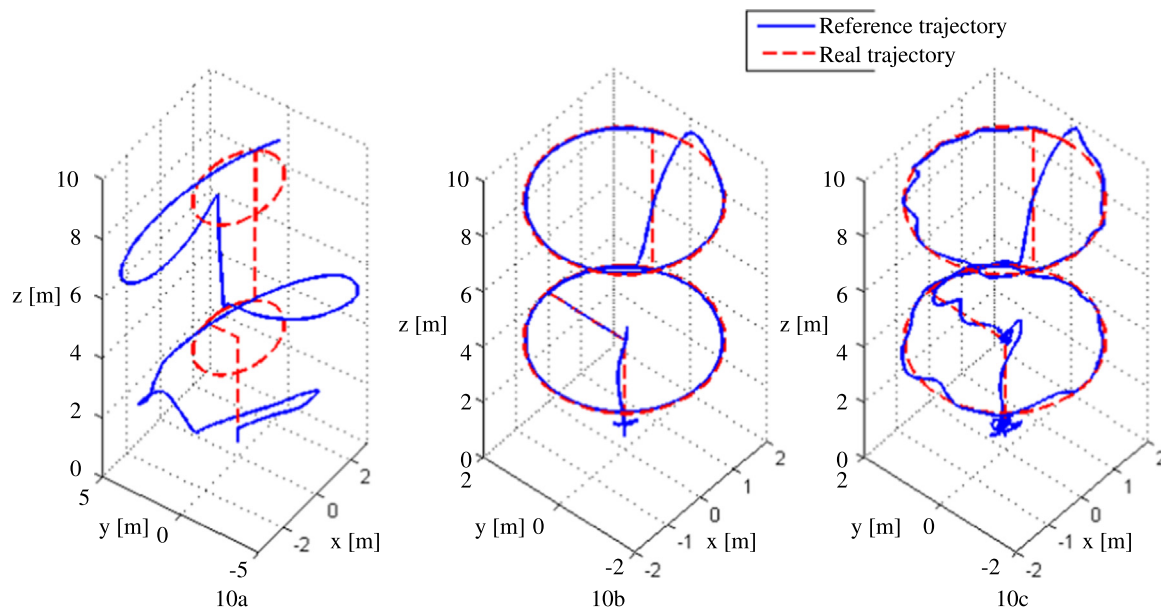


Fig. 10. Case d: Absolute position of the coaxial rotor.

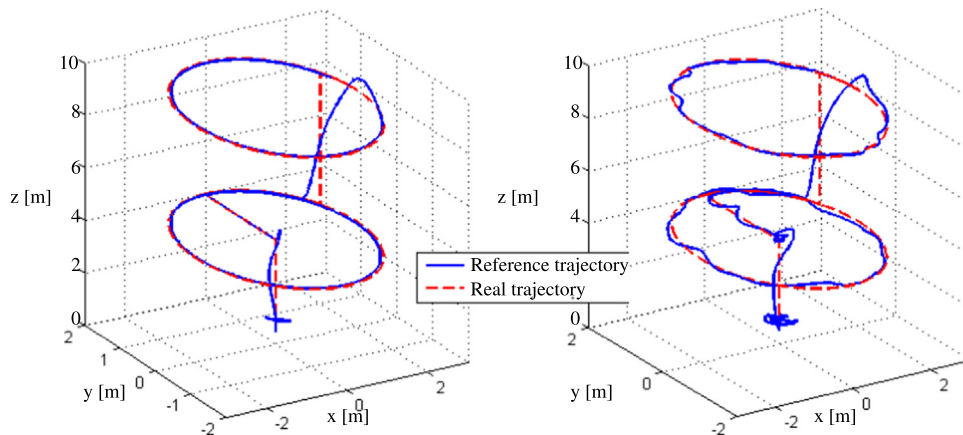


Fig. 11. Case d: comparison between the proposed controller and the adaptive backstepping approach [24].

4.5. Case e: comparison with the adaptive backstepping

The efficiency of the proposed control method is illustrated in Fig. 11 via the comparison between the absolute position obtained using the proposed controller and the adaptive backstepping presented in [24] in the presence of uncertainties and external disturbances presented. The simulation results shown in Fig. 11 clearly demonstrate better performances, faster convergence, high-precision tracking and robustness with respect to the adaptive backstepping controller presented in [24].

5. Conclusions

This paper presented a solution to the problem of tracking control of an underactuated coaxial-rotor vehicle along a predefined trajectory in the presence of unknown aerodynamic efforts. A complete dynamics model is designed in the case of quasi-stationary flight. Indeed, we have proposed an original control strategy based on hierarchical backstepping technique integrated with a new disturbance observer with finite time convergence (FTDO) to improve the performance of stabilisation and tracking control of a small CRUAV with bounded unknown aerodynamic efforts. Based on the nonlinear hierarchical technique, the overall system has been structured as an upper

subsystem describing the translational dynamics in cascade with a lower subsystem describing the orientation system, while the FTDO is constructed to estimate the unknown aerodynamic disturbances acting on the system. The controller has demonstrated good performances and stability of the closed-loop system.

As future work, it would be interesting to design a new mechanism for the small rotorcraft UAV in the case of aggressive flight. Also, it would be interesting to improve the control system by considering the aerodynamic effects at higher speeds and aggressive maneuvers. Further, the design of a nonlinear observer in order to estimate the unknown state (absence of sensor information).

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